

Integral of the convolution of two functions: $\int_{-\infty}^{\infty} (f * g)(x) dx$

Asked 9 years, 10 months ago Modified 9 years, 9 months ago Viewed 10k times



11



There is this proof for the integral of convolution between two functions:

$$\begin{aligned} \int_{-\infty}^{\infty} (f * g)(x) dx &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(x - \xi) g(\xi) d\xi \right] dx \\ &= \int_{-\infty}^{\infty} g(\xi) \left[\int_{-\infty}^{\infty} f(x - \xi) dx \right] d\xi \\ &= \int_{-\infty}^{\infty} g(\xi) \left[\int_{-\infty}^{\infty} f(\eta) d\eta \right] d\xi \\ &= \int_{-\infty}^{\infty} g(\xi) d\xi \int_{-\infty}^{\infty} f(\eta) d\eta \end{aligned}$$

What confuses me is the way author has easily changed the order of terms under integral sign. I'll appreciate any explanation.

[calculus](#) [integration](#) [functional-analysis](#) [convergence-divergence](#) [convolution](#)

Share Cite Edit Follow Flag

edited Jan 18, 2015 at 21:37

asked Dec 22, 2014 at 20:10



Olivier Oloa

122k

19

205

325



keyanm

113

1

7

1 Please what are the hypotheses on f, g ? – Olivier Oloa Dec 22, 2014 at 20:17

They are both continuous, piecewise smooth, and absolutely integrable functions. – keyanm Dec 22, 2014 at 20:21

1 Ok, so they are in $L^1(\mathbb{R})$. You may use Fubini-Tonelli Theorem. – Olivier Oloa Dec 22, 2014 at 20:22

1 Thanks a lot, I'm still trying to figure out how you did a similar argument for integration of F with respect to ξ . – keyanm Dec 22, 2014 at 23:42

You may notice that, by an easy change of variable $x - \xi \rightarrow u$, you get that $f * g = g * f$, so you can conclude concerning F with respect to ξ . – Olivier Oloa Dec 22, 2014 at 23:47

1 Answer

Sorted by: Highest score (default)



12



Hint. You may use [Fubini-Tonelli theorem](#).

Let $f, g \in L^1(\mathbb{R})$. Set $F(x, \xi) := f(x - \xi)g(\xi)$. Then

$$\int_{\mathbb{R}} |F(x, \xi)| dx = |g(\xi)| \int_{\mathbb{R}} |f(x - \xi)| dx = |g(\xi)| \int_{\mathbb{R}} |f(x)| dx = |g(\xi)| \|f\|_1$$

giving

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} |F(x, \xi)| d\xi \right) d\xi = \int_{\mathbb{R}} \|f\|_1 |g(\xi)| d\xi = \|f\|_1 \|g\|_1 < \infty.$$

In the same manner,

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} |F(x, \xi)| dx \right) d\xi = \|f\|_1 \|g\|_1 < \infty$$

Thus using Fubini-Tonelli theorem, we have $F \in L^1(\mathbb{R} \times \mathbb{R})$ and

$$\int_{\mathbb{R} \times \mathbb{R}} |F(x, \xi)| d(x, \xi) = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} |F(x, \xi)| dx \right) d\xi = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} |F(x, \xi)| d\xi \right) dx$$

Moreover, $\|f * g\|_1 \leq \|f\|_1 \|g\|_1$, since

$$\|f * g\|_1 = \int_{\mathbb{R}} |(f * g)(x)| dx = \int_{\mathbb{R}} \left| \int_{\mathbb{R}} F(x, \xi) d\xi \right| dx \leq \int_{\mathbb{R}} \left(\int_{\mathbb{R}} |F(x, \xi)| d\xi \right) dx = \|f\|_1 \|g\|_1.$$

Share Cite Edit Follow Flag

edited Dec 23, 2014 at 14:33

answered Dec 22, 2014 at 21:02



Olivier Oloa

122k

19

205

325